7. E-Commerce Customer Value & Purchase Prediction

Problem Statement

An e-commerce company has large volumes of customer and transaction data but lacks an integrated view of purchasing behavior and future customer activity.

Project Objective

Create a database-backed analytics and machine-learning system that identifies purchasing patterns and predicts a defined customer outcome.

Key Features / Scope

The project should include:

Database layer

- Customers
- Orders
- Products
- Categories
- Payments

SQL analytics

- Customer purchase frequency
- Revenue
- Average order value
- Product performance
- Repeat-purchase patterns

Python analytics

- Cleaning
- Missing-value treatment
- Feature creation
- EDA
- Correlation analysis

ML

Choose one:

- Purchase classification
- Customer-value regression
- Repeat-purchase prediction

Dashboard

- Revenue
- Customers
- Products
- Customer segments
- Model predictions

Expected Deliverable

A complete **E-Commerce Customer Intelligence Platform**.

Optional Extension

Implement experiment tracking with MLflow and expose the prediction model through FastAPI.